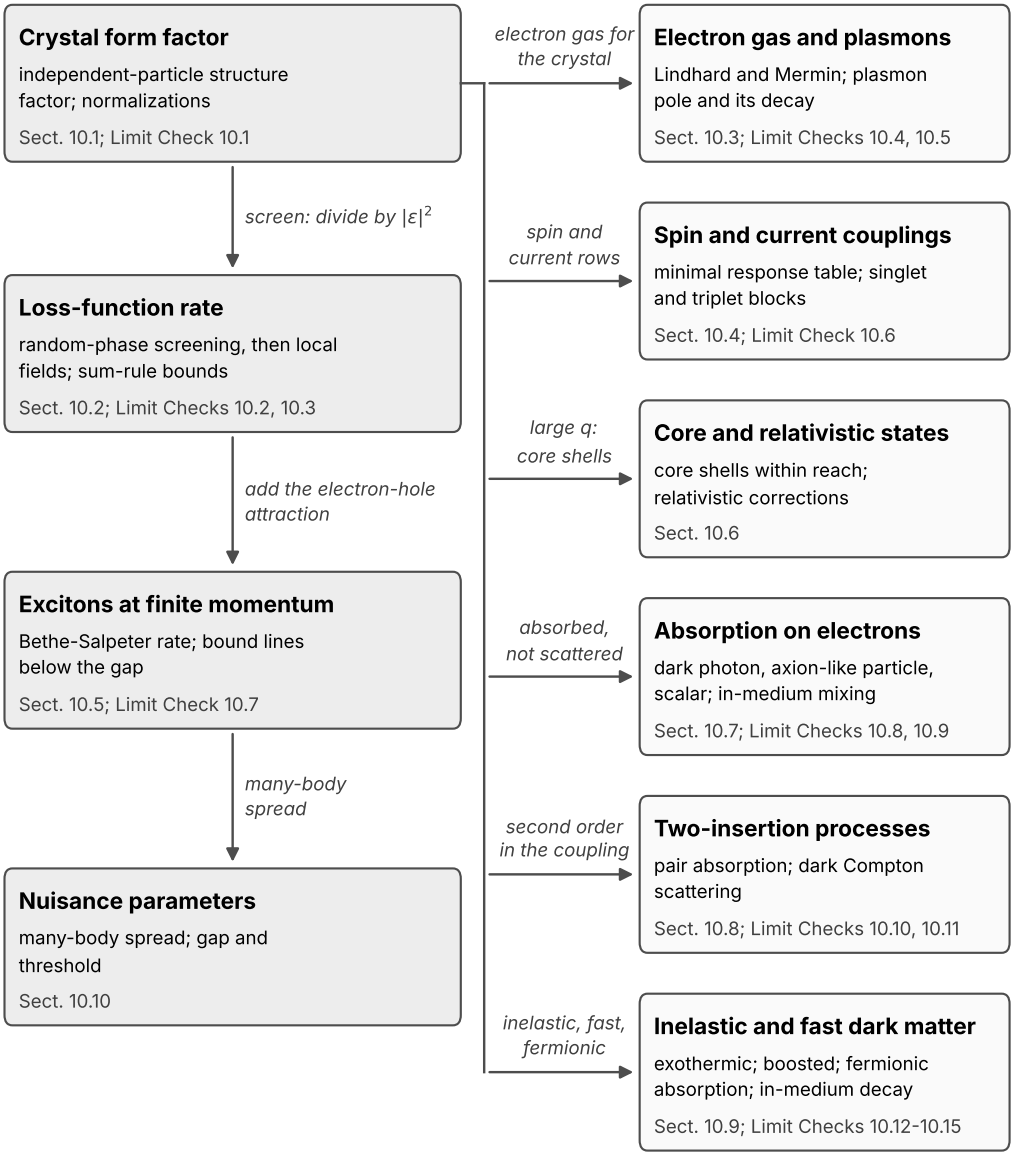
Crystal form factor
independent-particle structure factor; normalizations
Sect. 10.1; Limit Check 10.1
screen: divide by $|\varepsilon|^2$
Loss-function rate
random-phase screening, then local fields; sum-rule bounds
Sect. 10.2; Limit Checks 10.2, 10.3
add the electron-hole attraction
Excitons at finite momentum
Bethe-Salpeter rate; bound lines below the gap
Sect. 10.5; Limit Check 10.7
many-body spread
Nuisance parameters
many-body spread; gap and threshold
Sect. 10.10
electron gas for the crystal
Electron gas and plasmons
Lindhard and Mermin; plasmon pole and its decay
Sect. 10.3; Limit Checks 10.4, 10.5
spin and current rows
Spin and current couplings
minimal response table; singlet and triplet blocks
Sect. 10.4; Limit Check 10.6
large q: core shells
Core and relativistic states
core shells within reach; relativistic corrections
Sect. 10.6
absorbed, not scattered
Absorption on electrons
dark photon, axion-like particle, scalar; in-medium mixing
Sect. 10.7; Limit Checks 10.8, 10.9
second order in the coupling
Two-insertion processes
pair absorption; dark Compton scattering
Sect. 10.8; Limit Checks 10.10, 10.11
inelastic, fast, fermionic
Inelastic and fast dark matter
exothermic; boosted; fermionic absorption; in-medium decay
Sect. 10.9; Limit Checks 10.12-10.15